ChainShield AI

An Agentic Supply Chain Intelligence Platform for Shipment Risk Assessment, Disaster Information Analysis and Mitigation Planning

A. Project Title

ChainShield AI: An Agentic Supply Chain Intelligence Platform Integrating Machine Learning, Deep Learning, Transformer-Based Natural Language Processing, and Retrieval-Augmented Generation for Supply Chain Risk Assessment and Decision Support

B. Abstract

Background and Problem Statement

Supply chains are affected by shipment delays and external disruptions, including floods, earthquakes, severe storms and damage to transport infrastructure. Operations teams must assess the reliability of individual orders while reviewing incident reports that may contain information about damaged facilities, interrupted services or urgent response activities. These inputs are often examined separately, limiting the ability to obtain a timely and consistent operational assessment.

Traditional reporting tools can summarize historical shipment performance but cannot readily interpret unstructured incident messages. Conversely, text-classification systems can categorize disaster information but do not explain historical shipment risk or help an operations team assess practical follow-up actions. This project addresses the opportunity to bring these capabilities together in a single decision-support application.

Motivation and Significance

ChainShield AI aims to demonstrate how predictive analytics and language understanding can support the work of a supply chain operations team. Its objective is not simply to recognize a named disaster; it is to identify potentially useful incident reports from large volumes of text, distinguish the nature of information being reported, and make relevant evidence available for further operational review.

For example, when a report describes bridge damage following an earthquake, the system should distinguish that information from a general expression of sympathy, identify it as an infrastructure-related report, and recommend verification of the affected transport arrangements. Alongside this incident investigation, an operations manager may examine the historical dispatch-risk estimate for a selected order. The two assessments will be presented together without assuming that the historical order was actually affected by the reported event.

Proposed Solution and Approach

The proposed system will be developed as an academic prototype with eight integrated capabilities:

1. **Predictive Shipment Risk Assessment:** Historical order records will be used to train Logistic Regression and XGBoost models that estimate the probability of a recorded shipment late-risk outcome.
2. **Intelligent Incident Report Screening:** A Bidirectional Long Short-Term Memory (BiLSTM) model will screen disaster-related messages for relevance or informativeness, helping separate reports requiring attention from routine or low-information content.
3. **Transformer-Based Incident Information Analysis:** A fine-tuned DistilBERT model will identify supported disaster types and classify reported humanitarian information, including infrastructure and utilities damage, rescue activity and other classes present in the training data.

4. **Retrieval-Augmented Generation:** A curated document collection, semantic embeddings and FAISS-based retrieval will provide supporting operational guidance with references to the retrieved sources.
5. **Multi-Agent Investigation:** LangGraph will coordinate an Intelligence Agent, a Knowledge Agent and a Planning Agent to assess model findings, retrieve relevant information and prepare recommended follow-up actions.
6. **Scenario-Based Mitigation Planning*:** The application will compare hypothetical responses against clearly defined rules concerning cost, operational approvals and available actions; it will not claim to optimize real transportation routes.
7. **Explainable AI and Generative Reporting*:** Model interpretation techniques and a language model will prepare clear reports that distinguish historical patterns, classified incident content, supporting evidence and assumptions.
8. **Interactive Dashboard:** A Streamlit and Plotly application will present shipment-risk estimates, incident review queues, supporting evidence, model comparisons, alternative actions and investigation summaries.

** The features marked with an asterisk are proposed extensions to the core project scope. Their extent of implementation will be determined following initial feasibility assessment and team review, taking into account the available project timeline, data suitability and progress on the core deliverables. Any adjustments will be reflected in the subsequent project documentation and final submission.*

Datasets and Implementation Scope

Three publicly available research datasets will support the prototype. The DataCo Smart Supply Chain dataset will provide historical shipment and order data for predictive modeling. The India–Nepal Disaster Tweet Classification Corpus will support disaster relevance and flood/earthquake text classification. The CrisisMMD dataset will be used in its text-only form to investigate crisis-message informativeness and humanitarian-information categories, including infrastructure and utilities damage. No image processing is planned.

The classification tasks have different label definitions; therefore, records from the disaster datasets will not be combined indiscriminately. Each model will be trained and evaluated against the labels appropriate to its task. The final deployment may use separate classification models or compatible prediction components depending on experimental results and implementation feasibility.

A small, curated collection of publicly available guidance and project-authored procedures will support the retrieval component. Hypothetical operational constraints, such as approval requirements and illustrative costs, will be defined explicitly for the scenario-planning demonstration. Neither disaster dataset contains verified links to individual DataCo orders, and the prototype will not present such links as established facts.

Illustrative Business Use Cases

- Order Risk Prioritization: An operations manager selects historical orders and identifies those with higher model-estimated dispatch risk for further review.
- Infrastructure Incident Triage: The system screens incoming reports and highlights messages classified as describing infrastructure or utilities damage that may warrant verification.
- Disaster Information Review: Reports relating to floods or earthquakes are organized by their supported disaster and information categories, with uncertain results referred for manual review.
- Operational Investigation and Mitigation: An operator examines an order alongside a separately supplied incident report, retrieves relevant guidance and compares permitted follow-up actions under defined constraints.
- Incident Review Dashboard: A supervisor reviews the distribution of informative reports, infrastructure-related messages, classification uncertainty and outstanding investigation items.

Evaluation Methodology

Shipment-risk models will be assessed using chronological, order-level train/test separation and relevant measures such as precision, recall, F1-score and ROC-AUC. Features revealing shipping outcomes will be excluded from prediction inputs. For text analysis, the project will compare conventional TF-IDF methods with BiLSTM and DistilBERT where their tasks are comparable, reporting macro-F1, class-wise recall and confusion matrices. Special attention will be given to rare humanitarian-information classes and to errors passed from screening to downstream classification.

The retrieval and agentic components will be evaluated for document relevance, citation correctness, workflow completion, adherence to defined scenario constraints and the ability to communicate uncertainty. The final results will distinguish experimentally measured performance from proposed targets and illustrative demonstration values.

Expected Outcomes

- A trained and evaluated shipment late-risk prediction module using historical order data.
- A text-only natural-disaster intelligence module for incident screening and supported information classification.
- Comparative experimental results for Machine Learning, recurrent Deep Learning and transformer-based NLP techniques.
- An evidence-grounded information retrieval and question-answering facility.
- A functioning three-agent investigation workflow with validated, rule-constrained recommendations.
- An interactive dashboard and documented end-to-end demonstration scenarios.
- A final project report presenting results, limitations and opportunities for future expansion.

The expected academic contribution is a reproducible demonstration of how structured-data prediction, disaster-message understanding, retrieval-based evidence and coordinated AI agents can support more informed and transparent operational investigation.

C. Roles and Responsibilities

The project is organized into five complementary workstreams, with collaborative ownership across the team. The primary contributors listed for each workstream share responsibility for planning and delivery, while supporting contributors provide additional expertise, coordination and integration support. All members participate in the overall project outcomes.

Note: The workstream allocations and contributor associations presented below are indicative at this initial stage of the project. Members have expressed preferences based on their areas of interest and anticipated contributions. As the team progresses through dataset assessment, technical exploration and implementation planning, these allocations may be refined collaboratively to reflect evolving interests, project requirements, availability and workload. Confirmed responsibilities and ownership will be documented in the next applicable project submission(IF any) or the final report, as required by the institute.

Workstream 1 — Data Engineering and Machine Learning

Primary contributors: Sumojit Nath | Supporting contributors: Arnav Jain, Ankit Sood

Focus: Data Engineering and Machine Learning

Responsibilities:

- Acquire, prepare and validate the DataCo Smart Supply Chain dataset, including order-level aggregation and feature selection.
- Develop Logistic Regression and XGBoost shipment-risk models and evaluate them using appropriate validation methods.
- Investigate potential data leakage and produce interpretable model findings.
- Provide a reusable risk-prediction interface for the dashboard and Intelligence Agent.

Expected Contributions:

- Validated structured-data pipeline and dataset documentation.
- Trained ML models, evaluation results and explainability outputs.
- Integrated shipment-risk prediction service.

Workstream 2 — Deep Learning and Transformer-Based NLP

Primary contributors: Shivansh Sharma and Ankit Sood | Supporting contributors: Sumojit Nath

Focus: Deep Learning and Natural Language Processing

Responsibilities:

- Inspect and prepare the India–Nepal / CrisisMMD text datasets, documenting their actual labels and class distributions.
- Develop BiLSTM-based message screening for a clearly defined relevance or informativeness target.
- Fine-tune DistilBERT for supported disaster-type and/or humanitarian-information classification tasks.
- Compare neural and transformer results with a suitable TF-IDF baseline and assess minority-class performance.
- Provide reusable incident-screening and classification interfaces for downstream integration.

Expected Contributions:

- Prepared and documented disaster-text datasets.
- Trained BiLSTM and DistilBERT models with comparative evaluation.
- Incident-intelligence inference services and error analysis.

Workstream 3 — Retrieval-Augmented Generation and Knowledge Engineering

Primary contributors: Gokul S and Ankit Sood | Supporting contributors: Shivansh Sharma and Muthu Manickam

Focus: Retrieval-Augmented Generation and Knowledge Engineering

Responsibilities:

- Collect and organize reliable guidance concerning disaster response, infrastructure verification and supply-chain procedures.
- Develop document-processing, semantic embedding and FAISS retrieval components using LangChain.
- Provide source references and assess retrieval relevance and response grounding.
- Integrate knowledge retrieval with the LangGraph Knowledge Agent.

Expected Contributions:

- Curated, indexed knowledge collection.
- Working RAG service with source references.
- Retrieval and evidence-quality evaluation.

Workstream 4 — Agentic AI and Generative AI

Primary contributors: Shivansh Sharma and Muthu Manickam | Supporting contributors: Gokul

Focus: Agentic AI and Generative AI

Responsibilities:

- Design the Intelligence, Knowledge and Planning Agents using LangGraph.
- Integrate the shipment prediction, text classification and knowledge-retrieval interfaces.
- Connect an appropriate language model through OpenRouter, with a local option where feasible.
- Implement controlled tool routing, shared agent state, response validation and failure handling.
- Generate evidence-supported investigation reports and evaluate workflow reliability.

Expected Contributions:

- Integrated multi-agent workflow.
- Language-model integration and validated investigation reports.
- Agent execution logs and evaluation results.

Workstream 5 — Application Interface Development, Scenario Planning and Integration

Primary contributors: Arnav Jain | Supporting contributors: Shivansh

Focus: Application Development and System Integration

Responsibilities:

- Develop the interactive application using Streamlit and Plotly.
- Integrate shipment-risk analysis, incident classification, RAG results and multi-agent investigations.
- Implement historical analytics, review queues and constrained hypothetical scenario comparisons.
- Perform end-to-end testing, package the application and prepare deployment and demonstration instructions.

Expected Contributions:

- Integrated dashboard and scenario-analysis interface.
- Tested, reproducible application package.
- Final demonstration and deployment documentation.

Shared Team Responsibilities

All team members will contribute to literature review, dataset and software licensing compliance, architecture discussions, shared interfaces, integration testing, interpretation of experimental results, final report preparation and project presentation.

Project feasibility will be reviewed after the initial dataset audit. Where a specialized classification task is not adequately supported by the available text labels, it will be reported as a limitation or optional extension rather than presented as an established capability.